WidgetX Technical Operations Manual
Advanced Installation, Calibration, and Maintenance Procedures

1. Overview
This manual supplements the WidgetX Quick Start Guide with the detailed procedures required for industrial installation sites, including site preparation, sensor calibration, firmware maintenance, and troubleshooting of common fault codes. It is intended for certified field technicians and facility engineers responsible for keeping WidgetX units within their rated operating tolerances over a multi-year service life. Technicians must complete the WidgetX certification course before performing any procedure described in Sections 3 through 6 of this manual.

2. Site Preparation
Before installation, confirm that the mounting surface can support the unit weight of 42 kilograms plus a safety margin, that ambient temperature at the install location stays within 0 to 45 degrees Celsius year-round, and that a dedicated 20-amp circuit is available within 3 meters of the planned mounting point. Sites with ambient dust or particulate levels above the rated enclosure ingress protection rating require the optional sealed enclosure kit, ordered separately.

3. Electrical and Network Connections

Connect the unit to a grounded 20-amp circuit using the supplied power cable; do not use an extension cord rated below 20 amps, as undervoltage at startup can trigger a false thermal fault. Network connectivity is provided via the rear Ethernet port and should be connected to a switch port with Power over Ethernet disabled, since the WidgetX unit does not accept PoE and enabling it can damage the network interface. Once powered and connected, the unit performs a self-test sequence that takes approximately ninety seconds, after which the status LED should show solid green.

3.1 Common Fault Codes at Startup

Fault code E-01 indicates a grounding fault and requires an electrician to verify the circuit before retrying. Fault code E-02 indicates the ambient temperature sensor reading is out of range, most often because the unit was powered on before reaching thermal equilibrium after transport. Fault code E-07 indicates a network configuration error and typically resolves after re-running the network setup wizard from the front panel.

4. Pressure Sensor Calibration Procedure

4.1 Preparation

Before calibrating the pressure sensor, allow the unit to run for at least thirty minutes under normal operating load so that internal components reach a stable operating temperature; calibrating a cold unit produces readings that drift once the unit warms up. Gather the certified reference gauge, the calibration adapter fitting included in the technician toolkit, and a laptop running the WidgetX Configuration Utility version 4.2 or later, since earlier versions do not support the automated calibration wizard described below.

4.2 Calibration Steps

Attach the reference gauge to the calibration port using the supplied adapter, taking care to fully seat the fitting to avoid a slow leak that will produce an unstable reading during the hold phase. Launch the Configuration Utility, select Calibration from the main menu, and follow the on-screen prompts to apply reference pressure at the three calibration points: zero, mid-range, and full-scale. At each point, hold the pressure steady for thirty seconds while the utility samples the sensor output and computes the correction offset; do not proceed to the next point until the utility reports a stable reading, since an unstable sample will silently produce an incorrect offset that only surfaces as measurement drift weeks later in the field.

While the wizard samples each point, avoid touching the calibration adapter or the cable bundle running to the reference gauge, since even small mechanical vibration during the thirty-second hold can be picked up by the sensor and cause the utility to reject the sample and restart the hold timer for that point. On sites with significant floor vibration from nearby machinery, technicians are advised to perform calibration during a scheduled maintenance window when adjacent equipment is powered down, since repeated sample rejection can extend a routine calibration from its usual twenty minutes to well over an hour. The Configuration Utility logs every sample, accepted or rejected, along with a timestamp and the raw sensor voltage, which is useful when diagnosing an unusually long calibration session after the fact, and this log is bundled automatically into the calibration certificate described below so that a service center can review it if a unit is later returned under warranty.

Once all three points are captured, the utility calculates a linear correction curve and writes it to the sensor's calibration memory, and a calibration certificate is generated automatically for the maintenance log. Taking all of the above warm-up, attachment, and three-point sampling steps into account, the recommended calibration interval for the WidgetX pressure sensor under normal industrial use is 500 operating hours, after which drift outside the rated accuracy band becomes statistically likely.

5. Firmware Maintenance

Check the current firmware version from the Configuration Utility's System Information screen. Acme publishes firmware updates quarterly, and security-related patches are released out of cycle when needed. Before updating firmware, export the current calibration profile, since a firmware update resets calibration data to factory defaults and the unit must be recalibrated afterward using the procedure in Section 4. Firmware updates are applied through the Configuration Utility over the network connection and take approximately six minutes; do not power-cycle the unit during this window, as an interrupted update can require a factory reset by an authorized service center.

6. Preventive Maintenance Schedule

In addition to the pressure sensor calibration described in Section 4, technicians should inspect the intake filter every 250 operating hours and replace it if visibly clogged, and perform a full mechanical inspection of seals and fittings every 1,000 operating hours.

7. Troubleshooting Guide

If the unit reports intermittent pressure readings after the warranty period, first confirm the calibration interval in Section 4 has not been exceeded, since drift is the most common root cause of intermittent readings in units older than one year. If readings remain unstable immediately after a fresh calibration, inspect the calibration port fitting for a slow leak, and check that the reference gauge itself is within its own calibration due date, since a drifted reference gauge will silently miscalibrate the unit.

8. Warranty and Service

The WidgetX unit carries a two-year limited warranty covering manufacturing defects, excluding damage from improper installation (see Section 2 and Section 3) or from skipped calibration intervals. Contact an authorized service center for any repair; opening the sealed enclosure voids the warranty.